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Pilot Project of Surfactant-Polymer Injection to Enhance Oil Recovery from a High-Salinity Carbonate Reservoir in Arctic Conditions

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Summary

This paper presents the planning and implementation of a pilot surfactant-polymer (SP) flooding project in high-salinity carbonate reservoirs located in Arctic conditions. Two target formations – West Khosedayuskoe and Visovoye were selected. Both characterized by high salinity (>200 g/L), elevated temperature ($\sim 70^{\circ}\text{C}$), and mixed-wet to oil-wet rock surfaces.

A tailored surfactant-polymer formulation was developed through extensive laboratory testing and coreflood experiments. Field trials, including Single-Well Chemical Tracer Tests (SWCTT), demonstrated an increase in displacement efficiency by up to 17% and a decrease in residual oil saturation from 0.34 to 0.20.

The paper details the methodology for designing the surfactant-polymer pilot, including systematic laboratory screening, formulation optimization, and coreflood experiments.

Key stages of pilot field trials are structured. Project-specific risks—such as high adsorption, injectivity limitations, and operational constraints—are identified along with proposed mitigation strategies to enable successful field implementation.

These results indicate the technical feasibility and inform full-field implementation considerations.

Introduction

Carbonate reservoirs account for more than half of the world's remaining conventional oil reserves. However, their complex pore structure, low natural wettability, and pronounced heterogeneity pose significant challenges for enhanced oil recovery (EOR). These challenges are especially pronounced under harsh conditions such as high salinity and elevated formation temperatures. In recent years, chemical EOR—specifically surfactant-polymer flooding—has shown potential to improve displacement efficiency in carbonate settings.

Despite promising laboratory results, field-scale surfactant-polymer flooding applications in high-salinity carbonate reservoirs remain scarce. In Arctic regions, this gap is even more pronounced due to

logistical constraints, limited infrastructure, and low ambient temperatures that hinder chemical delivery and injectivity.

To address this challenge, a pilot surfactant-polymer flooding project was initiated in the Central-Khoreiver uplift region, targeting two oil-bearing formations: the West Khosedayuskoe and Visovoye fields. These formations exhibit extreme formation water salinity (>200 g/L), mixed- to oil-wet rock surfaces, and moderate formation temperatures ($\sim 70^{\circ}\text{C}$), as summarized in Table 1.

Table 1—General information on the considered fields of the Central-Khoreiver uplift

Parameter	West Khosedayuskoe field	Visovoye field
Target Formation	D ₃ fm-III	D ₃ fm-IV
Average Top depth (absolute reference), m	3065.0 (-2938.0)	3143.6
Reservoir Type	Layered-massive	Layered tectonically shielded
Pore System Type	Pore-cavernous-fractured	Pore-cavernous-fractured-cracked
Net Pay Thickness, m	7.7	8.0
Porosity, fraction	0.15	0.09
Initial Oil Saturation, fraction	0.83	0.87
Permeability (matrix), mD	3–4383 (average 554)	1–2685 (average 21)
Heterogeneity Coefficient, units	15	6
Initial Reservoir Temperature, $^{\circ}\text{C}$	70	69
Initial Reservoir Pressure, MPa	30.5	32.0
Oil Viscosity At Reservoir Conditions, mPa·s	11.6	6.6
Oil Density At Standard Conditions, kg/m ³	929	894
Bubble Point Pressure, MPa	11.01	8.0
Gas Oil Ratio (GOR), m ³ /t	43	39

The West Khosedayuskoe field is characterized by a cavernous matrix structure with permeability reaching up to 4 Darcy in some intervals (Yudin et al., 2017). Yet lacks a conventional pressure maintenance system. Water injection is conducted below the oil-water contact in select well pads. In contrast, the Visovoye field employs produced water reinjection directly into the productive formation.

The objective of this study is to demonstrate the technical feasibility of surfactant-polymer flooding in Arctic high-salinity carbonate reservoirs. A comprehensive methodology was implemented to assess displacement efficiency, injectivity, and operational risks under field conditions. This methodology included laboratory screening, formulation development, coreflooding, single-well tracer testing, and pilot field trials.

Laboratory Study and Formulation Design

The goal of the laboratory program was to design a robust surfactant-polymer formulation capable of maintaining stability. It also aimed to reduce interfacial tension (IFT) and provide favorable mobility control under extreme salinity and temperature conditions.

Surfactant-polymer flooding is considered a promising EOR method for these fields. The mechanism of action is based on the synergistic reduction of capillary and viscous forces, as described below.

According to the established theory, the effectiveness of oil displacement by chemical flooding is often characterized using the capillary number (N_c). This is a dimensionless parameter that quantifies the ratio of viscous to capillary forces (Cooke, 1971). The capillary number is calculated by Eq. 1.

$$N_c = \frac{\left(\frac{\Delta P}{L}\right) \cdot K}{\sigma} = \frac{g \cdot \mu}{\sigma}, \quad (1)$$

where g – velocity of the displacing phase flow in the porous medium, m/s; μ – viscosity of the displacing agent, Pa·s; σ – interfacial tension in the oil-displacing agent system, N/m; ΔP – pressure drop, Pa; L – sample length, m; K – permeability, m².

At low capillary numbers, a significant portion of oil remains trapped in pore throats due to capillary forces. Conversely, higher capillary numbers facilitate oil mobilization. A primary strategy to enhance the capillary number is to reduce interfacial tension using an aqueous surfactant solution.

The surfactant-polymer formulation was selected based on a combination of key performance indicators (Kornilov et al., 2019), obtained through screening and laboratory evaluation. The surfactant is a single-component anionic product produced by BASF, designed for stability under reservoir salinity conditions and temperature conditions. The polymer, also produced by BASF, was chosen based on an optimal price-to-viscosity performance ratio. The main properties of the selected surfactant-polymer formulation are summarized in Table 2.

Table 2—Main properties of the surfactant-polymer formulation

Parameter	Value
Interfacial Tension (Surfactant-Oil System), mN/m (209 g/L brine, 70°C, 10 kg/m ³ surfactant)	2–4·10 ⁻²
Static Surfactant Adsorption, mg/g of rock	9.5
Dynamic Surfactant Adsorption, mg/g of rock	0.5
Viscosity of Surfactant-Polymer Solution, mPa·s	5.5
Viscosity of Polymer Solution, mPa·s	4.1
Dynamic Polymer Adsorption, mg/g of rock	0.36
Irreversible Polymer Adsorption, mg/g of rock	0.05
Working Concentrations of Surfactant / Polymer, kg/m ³ / wt. %	10 / 0.12

The baseline implementation scenario proposed in the feasibility study involves the injection of 0.3 pore volumes (PV) of surfactant-polymer solution. This is followed by the injection of 0.7 pore volume of polymer solution. Subsequently, chase water flooding is carried out.

Coreflooding and Performance Evaluation

At the final stage of the laboratory program, coreflooding experiments were conducted to evaluate the displacement efficiency of the selected chemical formulation. These tests were carried out on various physical models, including bulk reservoir analogs, synthetic composite cores, and native core samples extracted from the target formations. The experiments aimed to assess the interactions between the formulated surfactant-polymer system, crude oil, and reservoir rock under conditions representative of the in-situ environment. Key parameters such as oil recovery, pressure drop, and effluent composition were monitored throughout the tests.

Fig. 1 presents the results of one such experiment. In this experiment, a surfactant slug was injected into a composite core consisting of four assembled core plugs. The total pore volume of the column was 45.8 cm³. Initial oil saturation corresponding to 37.3 cm³ of oil in place.

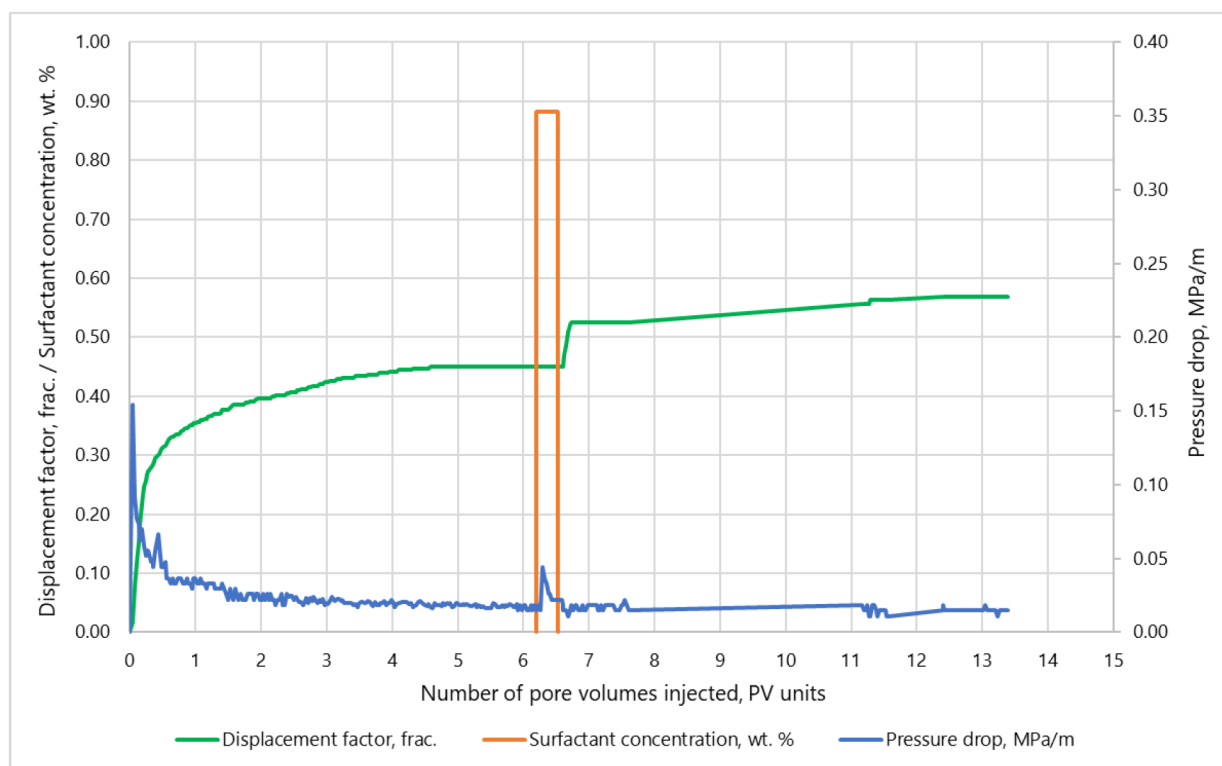


Figure 1—Evolution of Oil Recovery Factor, Surfactant Concentration, and Pressure Drop during Coreflood Testing with Surfactant Slug Injection (West Khosedayuskoe Field)

During the coreflood experiment, the oil recovery factor from baseline water flooding reached 0.450 (fractional units). Subsequent injection of a 0.3 pore volume surfactant slug at a concentration of 10 kg/m³ resulted in additional oil displacement. The incremental recovery factor after injecting one pore volume of fluid reached 0.075, and further increased to a final incremental value of 0.118. As a result, the total displacement factor achieved in the experiment was 0.568. This corresponds to a specific incremental oil recovery of 27.2 t of oil per tonne of injected surfactant. This confirms the economic and technical viability of the selected formulation.

Fig. 2 presents the results of a coreflood experiment involving a composite core assembled from four reservoir samples. The core has a total pore volume of 45.7 cm³ and an initial oil saturation corresponding to 37.1 cm³ of oil in place.

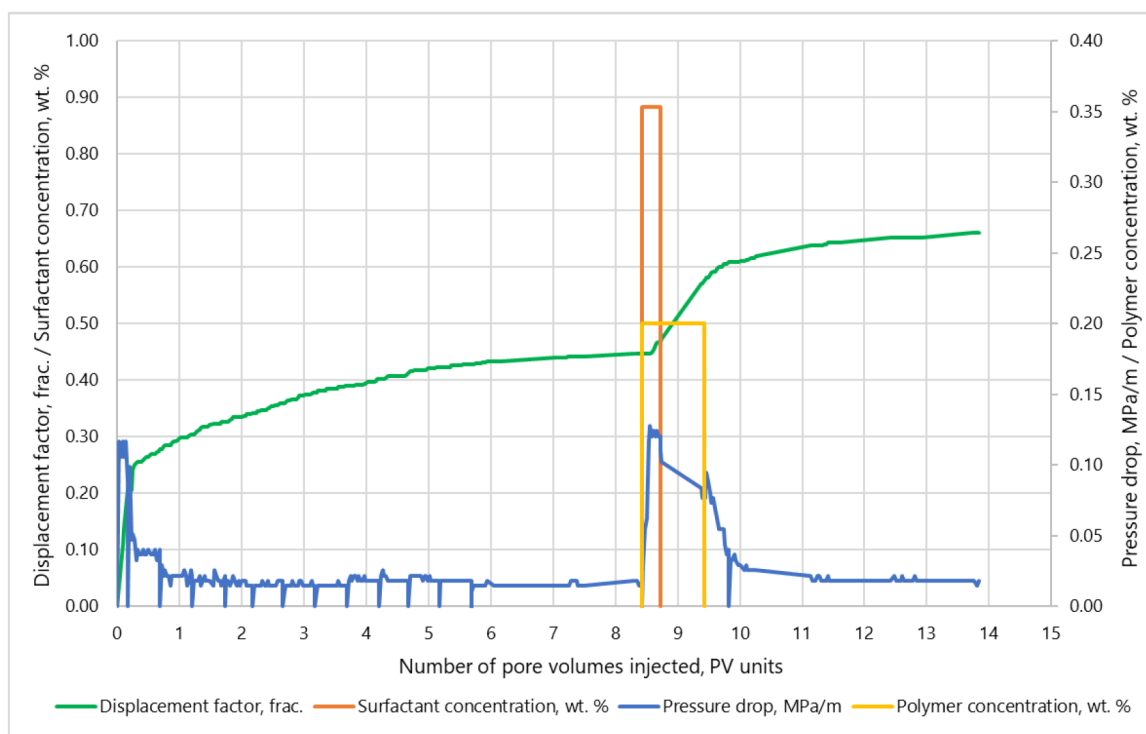


Figure 2—Improvement in Recovery Factor and Injectivity during Coreflood with Surfactant-Polymer Slug (West Khosedayuskoe Field)

In the baseline waterflooding phase, the oil recovery factor reached 0.446. A combined chemical flooding sequence was then performed, consisting of a 0.3 PV surfactant-polymer slug followed by a 0.7 PV polymer-only slug. The injected surfactant-polymer formulation had a surfactant concentration of 10 g/L and a polymer concentration of 0.2 wt.%. After injecting one pore volume of fluids, the incremental recovery factor reached 0.132, increasing to a final incremental recovery of 0.224. The total recovery factor at the end of the experiment was 0.670. The specific incremental oil recovery achieved by the surfactant-polymer system was calculated as 29.4 tonnes of oil per tonne of injected chemicals (t/t). This indicates high displacement efficiency under high-salinity and elevated-temperature conditions.

To construct a Capillary Desaturation Curve (CDC), a coreflood experiment was performed under varying capillary numbers. This involved using both water and surfactant-polymer injection at multiple flow rates. The goal was to quantify the variation in residual oil saturation (S_{or}) as a function of $\log(Nc)$.

Fig. 3 illustrates cumulative oil recovery from a composite core column as a function of injected pore volumes of water and surfactant-polymer solution. This experiment was designed to construct a capillary desaturation curve by quantifying the relationship between residual oil saturation and the logarithm of the capillary number. The test was conducted in two stages to expand the capillary number range over several orders of magnitude. In the first stage, water was injected at three increasing flow rates to raise the capillary number through changes in Darcy velocity. In the second stage, surfactant-polymer solution was injected at four increasing flow rates while simultaneously reducing interfacial tension by nearly three orders of magnitude. The combined variation of flow velocity and IFT enabled detailed assessment of oil mobilization under dynamic flow conditions. It also supported calibration of both laboratory-scale and field-scale recovery models.

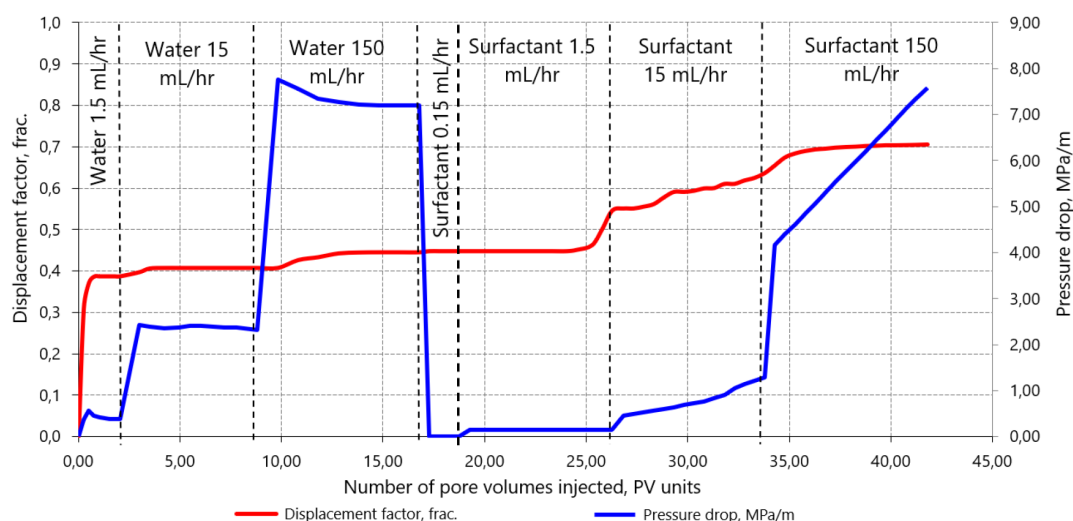


Figure 3—Coreflood-derived Capillary Desaturation Curve: Oil Recovery and Pressure Drop vs. Injected Pore Volumes under Varying N_c Conditions

The experimentally derived capillary desaturation curve describes the relationship between residual oil saturation and the logarithm of the capillary number. This curve was incorporated into a one-dimensional linear oil displacement model (Fig. 4). This model was also used as the basis for simulating the single-well chemical tracer test. Within the simulation framework, the absolute reduction in residual oil saturation between two sets of relative permeability curves—extracted from the CDC—was defined as $\Delta S_{or} = 0.272$. To align the model with the oil recovery dynamics observed in the coreflood experiment, the CDC input was calibrated. This adjustment accounted for differences in test conditions. The S_{or} -log(N_c) experiment had been performed under surfactant-rich, polymer-free conditions. These conditions do not fully represent the coupled effects present in the surfactant-polymer flooding scenario.

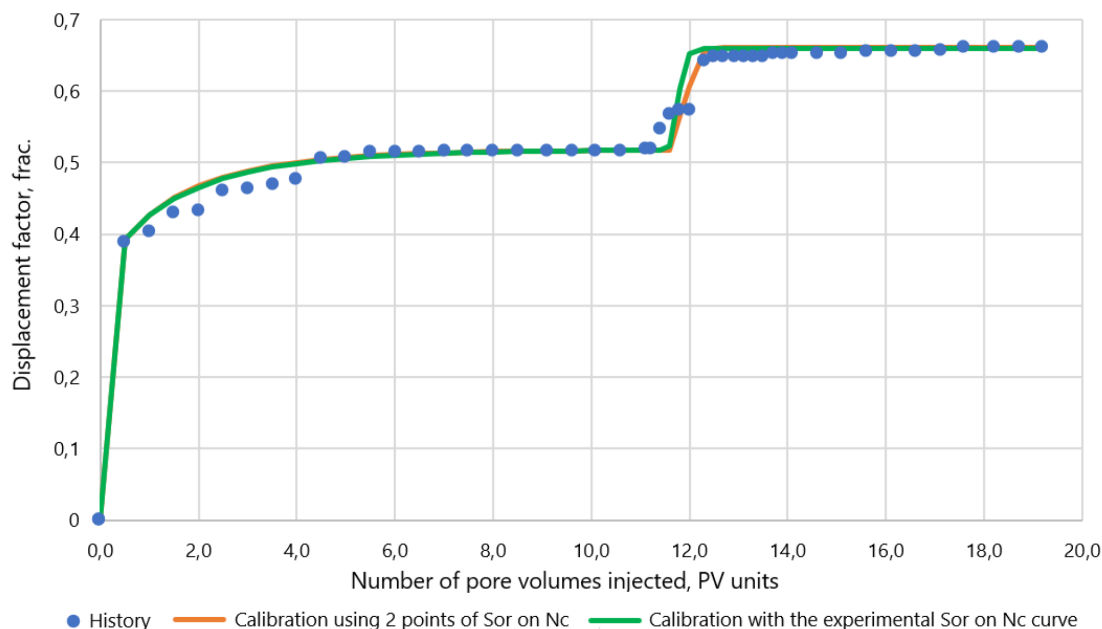


Figure 4—Comparison of Oil Recovery Curves for Different CDC in Coreflood Model Calibration

The calibrated S_{or} -log(N_c) relationship showed good agreement with coreflood recovery data. Table 3 compares model input parameters and oil recovery results across three scenarios: based on experimental

capillary desaturation curve, fixed S_{or} , and CDC-adapted surfactant-polymer flooding. The results demonstrate the impact of capillary-controlled mobilization on displacement performance.

Table 3—Comparison of Model Input Parameters and Oil Recovery Outcomes for Different S_{or} -log(N_c) Calibration Scenarios

Parameter	Experiment-Based CDC (S_{or} -log(N_c))	Fixed S_{or} (SP Only)	CDC-Adapted SP Flooding
Initial/Final S_{or} (S_{or1}/S_{or2})	0.53 / 0.25	0.42 / 0.29	0.42 / 0.148
log(N_c) ₁ /log(N_c) ₂	-7 / -0.3	-6.7 / -4	-7 / -0.3
ΔS_{or}	0.272	0.125	0.272
Surfactant adsorption type	irreversible		
Final oil recovery factor increase	26.9	14.2*	14.2*

* Recovery evaluation stopped after 1 PV of post-SP water injection.

Field Pilot Results: Evaluation of Surfactant Performance via Single-Well Chemical Tracer Test at the West Khosedayuskoe Field

Several methods can be employed to evaluate the in-situ effectiveness of chemical enhanced oil recovery formulations. These include the single-well chemical tracer test (Cooke, 1971), the Partitioning Interwell Tracer Test (PITT) (Fathi et al., 2012), and multi-well pilot flooding trials. Among these, single-well chemical tracer test offers a cost-effective and time-efficient option, though it comes with limitations. The test measures the displacement efficiency of the injected formulation—specifically, the reduction in residual oil saturation. However, it does not account for sweep efficiency and therefore cannot directly determine the overall oil recovery factor. In EOR applications, single-well chemical tracer test effectiveness is assessed by comparing residual oil saturation before and after injection of the chemical agent. Prior to field implementation, laboratory experiments must be performed to determine the partition coefficient of the partitioning (decomposing) tracer. These experiments also measure the rate of hydrolysis under reservoir conditions (Hu Guo et al., 2015). The partition coefficient defines the equilibrium distribution of the tracer between oil and water phases. The hydrolysis rate determines how rapidly the reaction product forms within the reservoir.

Fig. 5 presents the key stages of the pilot field program conducted at the West Khosedayuskoe field. The program was designed to evaluate the efficiency of oil displacement using the selected surfactant formulation.

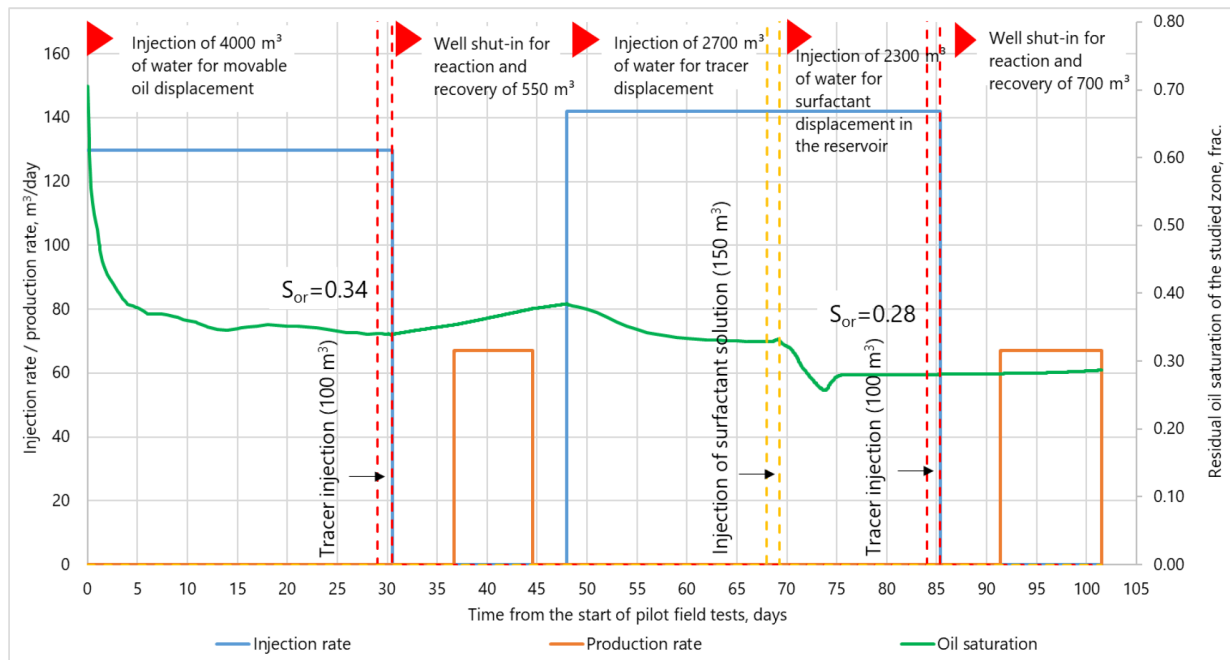


Figure 5—Evolution of Average Residual Oil Saturation (S_{or}) and Key Stages of the Pilot SWCTT Program at the West Khosedayuskoe Field

A simplified analytical approach for interpreting single-well chemical tracer test results has been widely adopted in global practice. This method relies on constructing a relationship between the concentration of produced tracers and the cumulative volume of fluid recovered during the sampling phase. The residual oil saturation is typically estimated using the following equation (Eq. 2):

$$S_{or} = \frac{(Q_a - \delta) / (Q_b - \delta) - 1}{(Q_a - \delta) / (Q_b - \delta) - 1 + K_d}, \quad (2)$$

where Q_a and Q_b – are the cumulative produced volumes corresponding to the peak concentrations of the partitioning tracer and its hydrolysis product, respectively (m^3); δ – wellbore volume, m^3 ; K_d – distribution coefficient.

However, Eq. 2 becomes inapplicable in complex reservoir conditions—such as heterogeneity, shut-in periods during sampling, dual-peak tracer curves, or operational anomalies. In such cases, the use of numerical hydrodynamic modeling with specialized software becomes essential for reliable interpretation.

Table 4 compares the interpretation results obtained using both the analytical method and hydrodynamic simulation. In both cases, the application of the surfactant formulation resulted in a 7% increase in displacement efficiency, confirming consistency between methods.

Table 4—Comparison of interpretation results of pilot field tests

Parameter	Analytical Interpretation	Hydrodynamic Modeling
Residual Oil Saturation (Waterflooding), frac.	0.31	0.34
Residual Oil Saturation (Surfactant Flooding), frac.	0.25	0.28
Displacement Factor (Waterflooding), frac.	0.65	0.61
Displacement Factor (Surfactant Flooding), frac.	0.72	0.68
Increment Displacement Factor, frac.	0.07	0.07

Field Pilot Results: Evaluation of Surfactant-Polymer Injection Efficiency at the Visovoye Field

Building on the experience gained from the pilot surfactant injection program at the West Khosedayuskoe field. A more advanced approach was adopted for the Visovoye field. In this case, hydrodynamic modeling was used not only for post-test interpretation but also as a tool for test design optimization. The modeling process enabled the selection of an optimal test configuration, allowing definition of key design parameters and operational targets in advance.

Fig. 6 presents the final single-well chemical tracer test design layout, developed through iterative scenario simulations using a validated reservoir model.

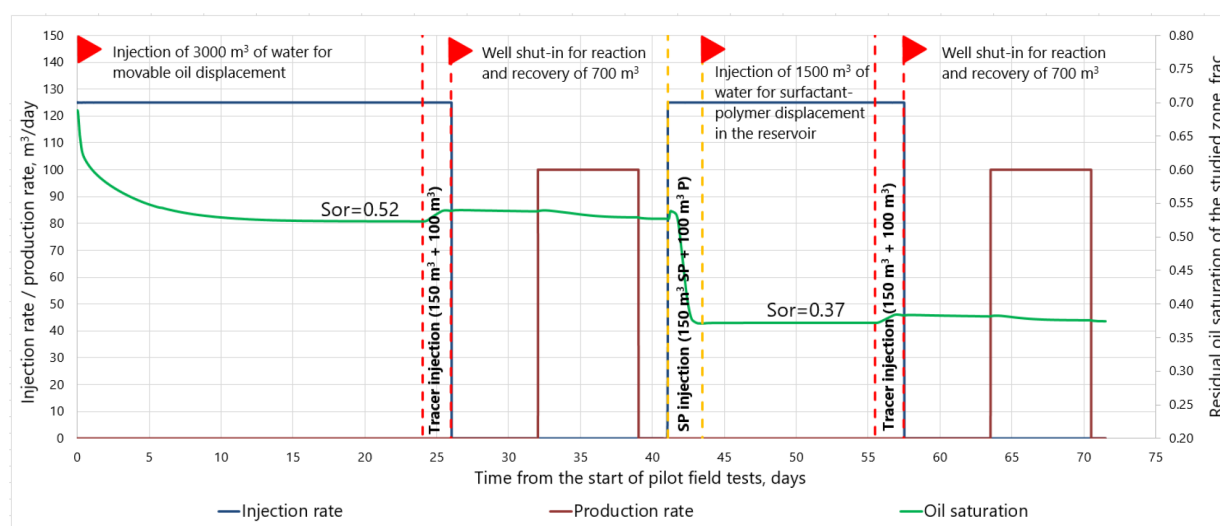


Figure 6—Evolution of Residual Oil Saturation (Sor) and Key Stages of the Pilot Surfactant-Polymer Injection Test at the Visovoye Field

Table 5 summarizes the key operational parameters and results of the pilot field test performed at the Visovoye field. The test was conducted to evaluate the efficiency of surfactant-polymer flooding. The test included sequential tracer studies (SWCTT 1 and 2), water preflush, chemical slug injection, and post-flush stages to assess incremental recovery.

Table 5—Key Parameters and Outcomes of the Pilot Surfactant-Polymer Flooding Test at the Visovoye Field

Parameter	Value
Injected Water Volume for Displacement of Mobile Oil, m ³	5300
Injected Surfactant-Polymer / Polymer Slug Volumes, m ³	150 / 150
Injected Water Volume for Displacing Surfactant-Polymer Slug into Reservoir, m ³	2300
Produced Fluid Volume During SWCTT 1 / SWCTT 2, m ³	720 / 710
Residual Oil Saturation after Waterflooding, frac.	0.34
Residual Oil Saturation after Surfactant-Polymer Flooding, frac.	0.20
Incremental Displacement Factor from Surfactant-Polymer Flooding, frac.	0.17

Fig. 7 shows the results of the single-well chemical tracer test adaptation conducted after the initial waterflood (SWCTT 1). The test interpretation was performed using a numerical hydrodynamic reservoir model, allowing for accurate matching of observed tracer behavior and residual oil saturation.

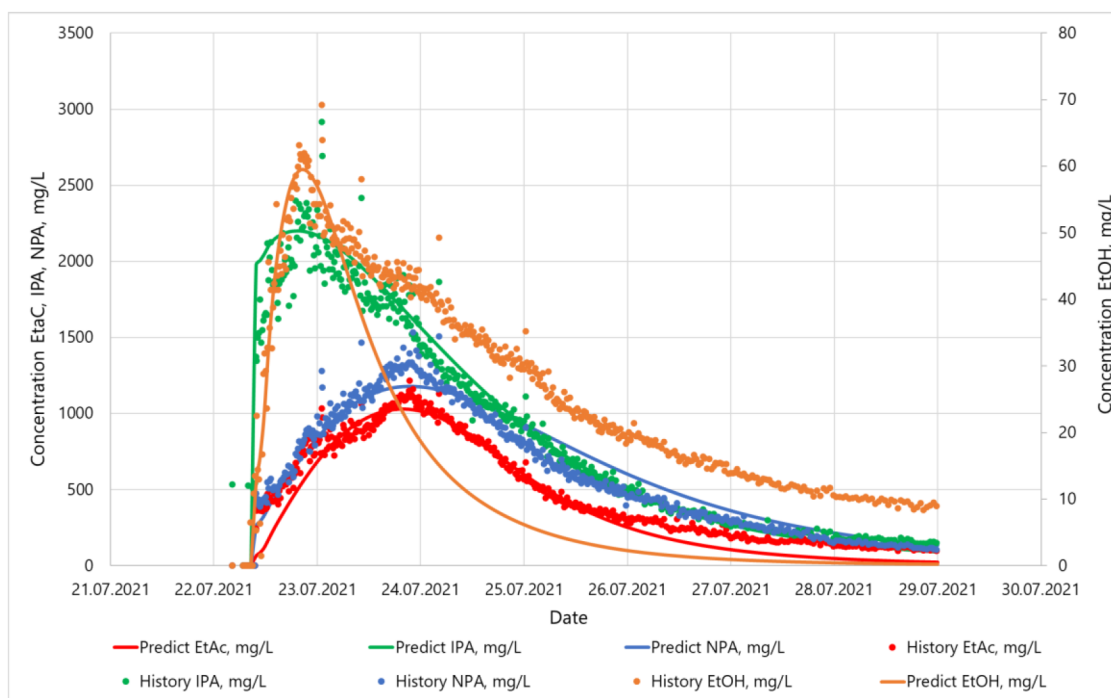


Figure 7—History Matching of Tracer Production Profiles Following Initial Water Injection (SWCTT 1) at the Visovoye Field

It is important to note that the observed mismatch between the simulated and measured EtOH concentration profiles may be attributed to insufficient sample preservation during field collection. Due to limited capacity of the on-site chemical laboratory, certain samples were not fully frozen. Freezing is required to halt the hydrolysis reaction of the decomposing tracer. As a result, continued reaction during storage could have led to overestimated concentrations of the hydrolysis product, thus distorting the tracer response interpretation.

Fig. 8 presents the history matching results for the second single-well chemical tracer test. This test was conducted after injection and displacement of the surfactant-polymer slug deeper into the reservoir.

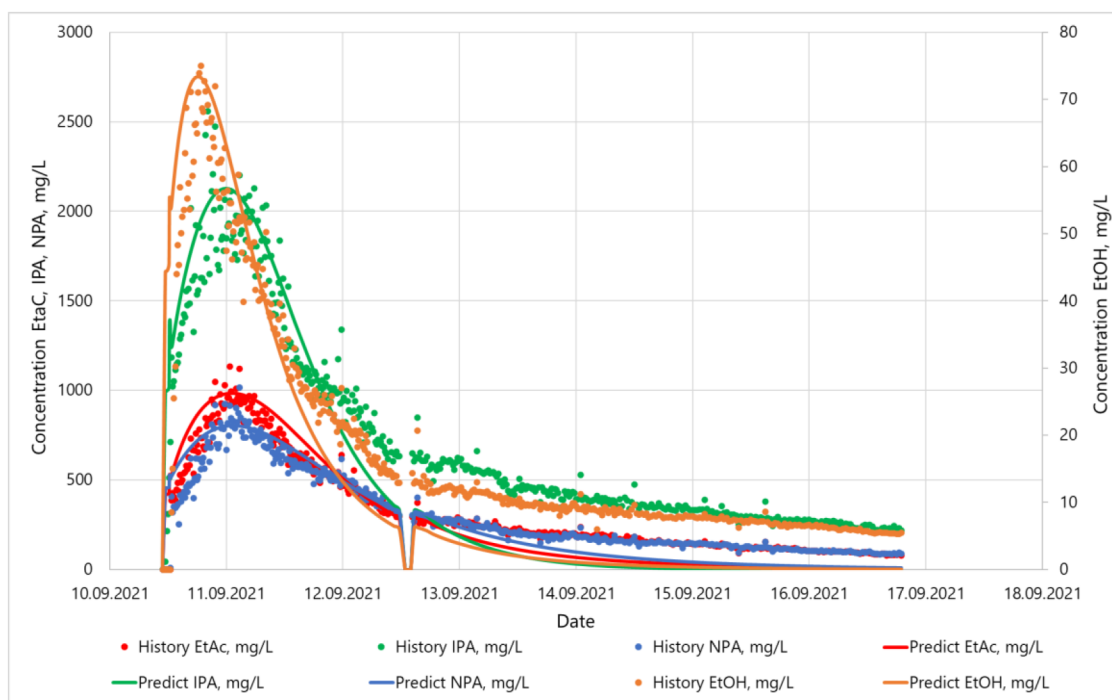


Figure 8—History Matching of Tracer Production Profiles Following Surfactant-Polymer Injection (SWCTT 2) at the Visovoye Field

In this case, the simulated concentration profile of the hydrolysis reaction product shows improved agreement with the measured field data. This indicates that the model adequately reproduces the key transport and reaction processes observed in the reservoir during the test. Once a satisfactory match between the modeled and observed tracer production profiles is achieved, the next step follows. It involves estimating the average oil saturation within the investigated zone of the model domain.

Single-well chemical tracer testing enables quantification of the volume-averaged oil saturation in the portion of the reservoir contacted by the injected tracers. For this region, the hydrodynamic model generates a curve showing the saturation averaged by pore volume over time. The confirmed effectiveness of the surfactant-polymer formulation was demonstrated by the single-well chemical tracer test results. This provides a technical basis for progressing to the next phase of project implementation. The next phase involves the pilot surfactant-polymer flooding within a multi-well injection pattern.

Pilot Project Planning and Execution Workflow

The primary objective of the pilot surfactant-polymer flooding project is to validate the operational feasibility and technical performance of the selected chemical formulation under reservoir conditions. The pilot aims to:

- De-risk full-field deployment by addressing key operational and reservoir uncertainties
- Assess the incremental displacement efficiency of the surfactant-polymer formulation under representative field conditions
- Evaluate well responses within the surrounding drainage area
- Refine injection and monitoring strategies for broader implementation.

Following the integration of results from the pilot and history matching of the hydrodynamic model, a comprehensive technical and economic justification is performed. This justification supports the upscaling of the surfactant-polymer flooding process to a full development unit.

Two candidate formations were considered for implementation of the surfactant-polymer flooding pilot project, with key reservoir and operational parameters summarized in Table 6. The selection was based on factors such as fluid flow infrastructure, well configuration, available pore volume, injectivity, and potential oil production losses during conversion of production wells to injectors.

Table 6—Parameters of reservoir sections for selecting the main candidate for the surfactant-polymer flooding pilot project

Object	Visovoye Field	West Khosedayuskoe Field
	Injection in D ₃ fm-IV	Injection in D ₃ fm-III
Underlying Aquifer	-	+
Existing Water Injection System	+	-
Planned Additional Drilling	-	+
Pore Volume, thousand m ³	~ 620	~ 240
Estimated Oil Production Loss due to Injector Conversion, thousand tons	60.8	11–80
Actual Injectivity, m ³ /day	150	135–228
Average Perforated Oil-Saturated Thickness, m	9.4	6.8
Distance to Nearest Ring Wells, m	600	300
Injection Rate, PV/year	0.1	0.5

Based on a comparative assessment of the candidate reservoir sections, the West Khosedayuskoe field was selected for pilot implementation. This choice was due to the potential for faster performance evaluation and greater scalability. These advantages are attributed to the existing well grid and lower pore volume of the targeted intervals.

A review of global pilot implementations (Table 7) (Hu Guo et al., 2017; Jabri et al., 2019; Kamal et al., 2017; Ling, 2015; Pratap et al., 2004) demonstrates the successful application of chemical EOR in reservoirs underlain by active aquifers. These precedents suggest that, despite the presence of a strong bottomwater drive in the D₃fm-III reservoir at West Khosedayuskoe, chemical injection remains technically viable. This is provided that optimized injection strategies and risk mitigation measures are applied.

To address potential risks associated with fracture-driven water breakthrough and reduced injection efficiency due to active bottomwater, a phased approach to project implementation is recommended:

1. Phase 1 – Waterflood Evaluation and Injection Pilot:

Conduct a pilot injection of underflow (produced) water by converting a central cell well to injection mode.

2. Phase 2 – Surfactant-Polymer Flooding:

Upon successful completion of Phase 1, proceed with surfactant-polymer flooding in adjacent reservoir sectors based on updated model calibration and field performance.

Throughout the project, continuous improvement of the surfactant-polymer formulation is being pursued to enhance performance and operational efficiency. Key areas of development include:

- Increasing specific displacement efficiency and robustness under field conditions
- Simplifying field-scale preparation and injection procedures
- Enhancing economic feasibility by reducing surfactant and polymer costs
- Minimizing logistics through higher active content and sourcing domestic chemical components.

Conclusion

1. Surfactant-polymer flooding has been evaluated as a viable enhanced oil recovery method for the Central-Khoreiver uplift fields. These fields are characterized by carbonate reservoirs with harsh conditions. The conditions include high formation water salinity (~ 200 g/L) and elevated reservoir temperatures ($\sim 70^\circ\text{C}$). A tailored surfactant-polymer formulation was selected that achieves ultra-low interfacial tension ($2\text{--}4 \times 10^{-2}$ mN/m) in the high-salinity oil-water system.
2. Laboratory coreflood experiments confirmed the high displacement efficiency of the selected formulation. Results from pilot field tests aligned with laboratory findings, demonstrating an incremental displacement factor of +14–17%. This validates the technical feasibility of surfactant-polymer flooding under Arctic carbonate conditions.
3. To accurately simulate surfactant-polymer injection performance, it is critical to incorporate the dependency of residual oil saturation on the capillary number. This requires multi-rate coreflood experiments with both brine and chemical solutions. It also involves the implementation of the derived Sor-log(N_c) relationship into reservoir models via calibration against partitioning tracer tests and full-field simulations. The resulting model adaptations showed satisfactory history matching.
4. Field selection criteria and pilot area configuration were established based on geological, operational, and economic factors. The West Khosedayuskoe field was selected for pilot implementation due to favorable pore volume, infrastructure, and scalability potential. Adjacent reservoir blocks were also identified for follow-up chemical injection trials under comparable conditions.

Nomenclature

CDC	Capillary Desaturation Curve
δ	wellbore volume, m^3
EtOH	ethanol
EOR	Enhanced Oil Recovery
K_d	distribution coefficient
K	permeability, m^2
L	sample length, m
μ	viscosity of the displacing agent, $\text{Pa}\cdot\text{s}$
N_c	capillary number
PITT	Partitioning Interwell Tracer Test
ΔP	pressure drop, Pa
PV	Pore Volume
Q_a	the cumulative produced volume corresponding to the peak concentrations of the partitioning tracer, m^3
Q_b	the cumulative produced volume corresponding to the peak concentrations of hydrolysis product, m^3
σ	interfacial tension (IFT) in the oil-displacing agent system, N/m
SP	Surfactant-Polymer
Sor	residual oil saturation
SWCTT	Single-Well Chemical Tracer Test
ϑ	velocity of the displacing phase flow in the porous medium, m/s

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